OCS INTELLIGENCE · SPRINT 4 · PLAYBOOK

Locking down NetBird with one-way group policies

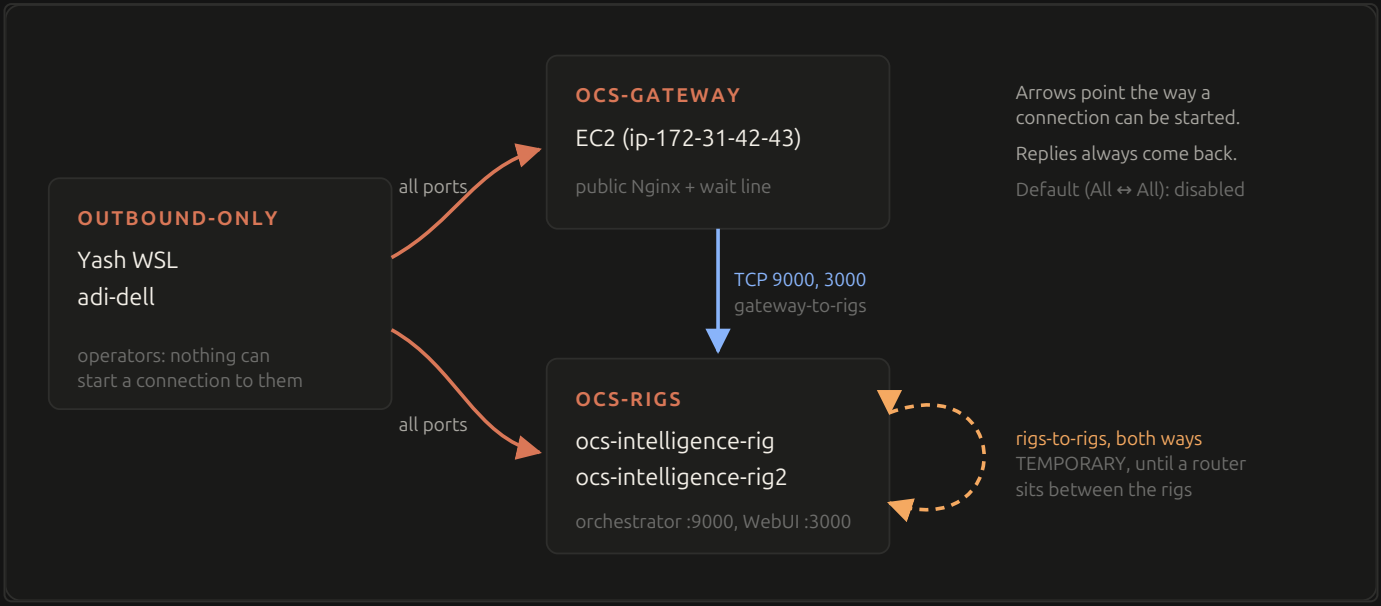
Servers stay reachable for operators and for each other, and nothing on the network can open a connection to an operator's own machine.

STATE **done** | DATE 2026-10-15 | BY Yash Parikh | SOURCE evidence/sprint4/2026-10-15-netbird-one-way-access.md

On 2026-10-15 we replaced NetBird's built-in `Default` policy, which let every peer connect to every other peer in both directions, with three group-based policies. Operator machines now sit in an `outbound-only` group: they can reach every server, but nothing on the network can start a connection to them. The EC2 gateway reaches the rigs only on the two ports it needs. Later the same day a temporary `rigs-to-rigs` rule was added so Open WebUI on Rig 1 could call the model on Rig 2. That link is what the Sprint 4 inference playbook depends on. The public API and operator SSH were checked before and after the change, and both kept working.

At a glance

Goal	No peer, including a compromised server, can open a connection to an operator's laptop or WSL
Outcome	Done. `Default` disabled; three one-way policies live; `apply.py check` matches `acl.json`
Where	The NetBird account (dashboard and management API). No host config changed
Desired state	`infra/netbird/acl.json`, checked with `infra/netbird/apply.py`
Why	Decision 0008, **proposed**, awaiting a second operator's review



Who can start a connection to whom after the change. Arrows point from the side that may start a connection; replies always come back.

Before

NetBird holds both the servers (EC2, Rig 1, Rig 2) and operators' personal machines. Every peer was in the single group `All`, and these were the policies:

POLICY	SOURCE → DESTINATION	DIRECTION	PORTS
`Default`	`All` → `All`	both ways	all
`SSH Access` (×3)	`All` → rig, rig2, EC2	one-way	NetBird SSH 22
Temporary access (×2)	a browser-client peer → rig	one-way	NetBird SSH 22022

`Default` was the only policy that let anything reach `Yash WSL` or `adi-dell`. Under it, a compromised rig could open connections into an operator's machine. The threat model also flagged that the rig's orchestrator port `9000` had no firewall rule, so every peer could reach it and skip the EC2 wait line.

What we did

NetBird policies run from a source group to a destination group, and the client firewall is stateful. A one-way policy lets the source start connections and lets the replies come back, but the destination can't start a connection of its own. That property is what the whole design rests on.

1. **Write the desired state down first.** `infra/netbird/acl.json` lists the groups, their members, and the policies. `apply.py` only compares and creates. It never deletes, never edits an existing policy, and never changes group membership (that's done in the dashboard).
2. **Get an API token from an admin service user** and put it in the repo-root `.env`, which is gitignored:

```
# .env  (template: .env.example)
NETBIRD_API_TOKEN=nbp_CHANGE_ME
```

3. **Record the "before" state.** These checks were run from `Yash WSL`:

```
https://ai.opencodingsociety.com/healthz 200
https://ai.opencodingsociety.com/ 200          # EC2 Nginx -> rig :3000 over NetBird
WSL -> 100.75.123.203:9000 open
WSL -> 100.75.123.203:22 open
WSL -> 100.75.172.167:22 open
```

4. **Create the groups and policies**, still with `Default` on, so nothing breaks yet:

```
python3 infra/netbird/apply.py check    # read-only: lists what's missing
python3 infra/netbird/apply.py apply    # creates missing groups and policies
```

GROUP	MEMBERS
`outbound-only`	`Yash WSL`, `adi-dell`
`ocs-gateway`	`ip-172-31-42-43` (EC2)
`ocs-rigs`	`ocs-intelligence-rig`, `ocs-intelligence-rig2`

POLICY	SOURCE → DESTINATION	DIRECTION	PORTS
`gateway-to-rigs`	`ocs-gateway` → `ocs-rigs`	one-way	TCP 9000, 3000
`outbound-only-to-infra`	`outbound-only` → `ocs-gateway`, `ocs-rigs`	one-way	all

5. **Disable `Default`** (it's kept, not deleted, so it can be switched back on):

```
python3 infra/netbird/apply.py default off     # "on" rolls back
```

6. **Re-run every check about 20 s later.** All the same results came back, plus `WSL -> 100.75.123.203:3000 open`. `netbird status -d` on `Yash WSL` then listed only the three servers. `adi-dell` and the browser-client peers had dropped out of its network map, as expected for two `outbound-only` peers.

7. **Amendment, same evening: `rigs-to-rigs`.** Open WebUI runs on Rig 1, and Sprint 4 needed it to call the model on Rig 2. We added a bidirectional `ocs-rigs` → `ocs-rigs` rule on all ports (commit `c2ac00d`). Restricting server-to-server traffic buys little here, because EC2, the most exposed host, already reaches both rigs. What 0008 protects is the `outbound-only` machines, and that rule didn't change.

> **WARNING**
>
> `rigs-to-rigs` is **temporary**. Remove it from `acl.json` and the dashboard once a router (the planned Mac Mini, or EC2) sits between the rigs.

How to check it

CHECK	PASS LOOKS LIKE	IF IT FAILS
`python3 infra/netbird/apply.py check`	`live NetBird config matches acl.json`	Someone changed policies in the dashboard. Decide which side is right, then fix the other
`curl -s -o /dev/null -w '%{http_code}' https://ai.opencodingsociety.com/healthz`	`200`	EC2 Nginx or the gateway is down. That isn't NetBird
`curl -s -o /dev/null -w '%{http_code}' https://ai.opencodingsociety.com/`	`200` (EC2 → rig `:3000` over `gateway-to-rigs`)	Check that `gateway-to-rigs` is enabled and EC2 is in `ocs-gateway`
`nc -vz -w5 100.75.123.203 9000` from an operator machine	`open`	Your peer isn't in `outbound-only`: dashboard → Peers → your peer → Groups
`netbird status -d` on an operator machine	Lists only servers, not other operators	Another operator peer is in a group that a policy points at
`python3 -m pytest -q tests/test_netbird_acl.py`	Passes. It asserts no policy has `outbound-only` as a destination	Someone made `acl.json` unsafe

Pitfalls

SYMPTOM	CAUSE	FIX
`403 permission denied` when creating a group	The API token belonged to a service user with the `user` role, which can read but not write	Use a token from a service user with the **admin** role
A new operator can SSH in but can't reach `:9000` or `:3000`	New peers join no group. They only get the `SSH Access` policies, whose source is `All`	Add the peer to `outbound-only` in the dashboard
Open WebUI on Rig 1 can't reach Rig 2's `:9000`	There's no policy between two members of `ocs-rigs` unless one says so	`rigs-to-rigs` (temporary)
A new service across NetBird doesn't answer	The default is now deny	Add a rule to `acl.json`, run `apply`, and update network

Open items

- **Second-operator review.** Blueprint §6 asks for two-person review of firewall changes. This change was applied live by one person, so 0008 stays `proposed` until a second operator reviews `acl.json` and runs `apply.py check`.
- **Not checked yet: a connection from a server to an operator machine.** We expect it to time out. Checking it needs a shell on a rig or on EC2.
- **Not checked yet: dashboard browser SSH after the change.**
- **Not checked yet: `/v1/*` end to end under the new policy.** It uses the same EC2 → rig `:9000` path, which needs a student key. `/` exercises the same rule on `:3000`.
- **Remove `rigs-to-rigs`** once a router sits between the rigs (roadmap).

Sources

- NetBird access policies: before and after, the verbatim checks
- Decision 0008 and its 2026-10-15 amendment
- Network, the current group table
- Commits `b8d12e4` (lockdown, `infra/netbird/`, tests) and `c2ac00d` (`rigs-to-rigs`)